

Testbench Architecture: Directed Sim + FPV

dotprod_int8_directed_tb.sv (sim path) + dotprod_top_sva (FPV path)

directed sim (dotprod_int8_directed_tb.sv)

Stimulus

initial block
6 directed cases:
zeros, min*max, max*max
cancel, single, ramp
a[8], b[8] 8b signed

a,b

dotprod_top

(DUT)

mode = FMT_INT8
front_end_int8 x8
mul_lane x8
align + acc_tree + round
combinational only

result, sat

Scoreboard

check() task
result == dotprod_ref?
sat == ref_sat?
\$error + \$fatal on mismatch

a,b (same inputs)

dotprod_ref

SV function (golden)
exact accumulate, sat_cast
single source bind truth
(DUT ports)

ref_r, ref_sat

PASS / FAIL

"DOTPROD_INT8_DIRECTED
PASS (6 cases)"

same dotprod_ref

FPV path (bind dotprod_top_sva + fpv_run_top.tcl)

dotprod_top_sva

bound via bind statement
am_int8_mode (assume)
a_result_matches_ref (assert)
c_sat_unreachable, c_pos, c_neg

properties

VC Formal FPV

check_fv -block
clock -none
reset -none
PROVEN / FAILED / UNREACHABLE